

“Modeling and simulation help to design a perfect combination of automotive subsystems”



At MathWorks' second annual India user conference in Bengaluru, customers and technical experts discussed the capabilities of MATLAB and Simulink for design, modeling and simulation.

Jalaja Ramanunni of Electronics For You had a chance to speak to Steve Miller, technical marketing manager, physical modeling, MathWorks, who explained how physical modeling capabilities enable automotive engineers to identify and fix problems early in design, rather than late in development when testing the final hardware

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What's the role of simulation in automotive?

Traditionally, when people develop systems, they use hardware. That means if you build a prototype, test it, and find that it does not work, you'll have to start all over again—build a new prototype and iterate on the design that way. This can be a long process. Engineers can now use simulation to make designs virtually before committing to hardware and prototyping.

Hardware prototyping has been used for a very long time. The first design starts with a prototype, and eventually, the designer comes up with a car that he is willing to sell. But this process takes a very long time—to come up with an idea, build it, test the hardware, iterate the idea and so on.

By using computers for modeling and simulation, designers can perform iterations faster. So feedback such as “The engine that took you two months to build requires changes. Make it ten percent smaller and we will see if it works for us” is painful to hear if you are developing using hardware prototypes. However, using a computer model, it takes much less time to make such large changes.

What are the challenges in automotive development? How does modeling and simulation help to address these?

One of the benefits of model-based design is that from

the initial stages, you can create a model of the system in the computer and test and verify all the requirements. By the time you build the first piece of hardware, say, the first car engine, you would have already discovered a lot of problems you typically find at a later stage when it is much more expensive to fix them.

When automotive companies develop a car, they focus on many aspects, cost-efficiency being one of the key areas. Getting to the market soon is also one of their focus areas. Since they are not the only company building cars, they would want to build a car as quickly as possible and release it in the market. Sometimes, it takes over six years to come up with a product. This is a risky situation for an automotive company as others might beat it.

Once an automotive company comes up with an idea, it should execute that idea immediately. If you can shorten the development process, you can save a considerable amount of money by building fewer prototypes and reducing costs in the development process.

A hand-built prototype can cost millions of dollars. In the first prototype, we often see that the manufacturing process is not yet in place and that the prototype can get insanely expensive. The more the engineer can do in a virtual environment, the less it will cost to develop the idea to a point where it can be put to production.